# Engineering Document E19 — Integration Platform, SDK & Marketplace Architecture

**Product Name:** HQ

**Engineering Package:** Version 1.0

---

## Purpose

This document defines the Integration Platform that enables HQ to connect securely with third-party services, enterprise systems, developer extensions, and future AI capabilities.

The objective is to make HQ an **AI Executive Operating System**, not just another AI application.

Every integration should behave like a native department within HQ.

---

## Vision

HQ should become the central intelligence layer for an organization.

Instead of users moving between dozens of applications, HQ coordinates them through a unified integration platform.

Examples:

- GitHub
- Slack
- Discord
- Gmail
- Google Workspace
- Microsoft 365
- Notion
- Jira
- Linear

- HubSpot
- Salesforce
- Stripe
- Shopify
- Figma
- Trello
- Asana
- Zapier
- Make.com
- Future Enterprise Systems

---

## Core Principles

The Integration Platform must be:

- Secure
- Modular
- Event-Driven
- Extensible
- Provider Independent
- Observable
- Versioned

Every integration follows the same engineering contract.

---

## Platform Architecture

```
External Service

↓

Connector

↓

OAuth Manager

↓

Secrets Vault
```

```
↓
Integration Gateway
↓
Workflow Engine
↓
Mission Orchestrator
↓
Executive Board
↓
Organization
```

Every integration enters HQ through the Integration Gateway.

---

# Integration Categories

## Communication

- Gmail
- Outlook
- Slack
- Microsoft Teams
- Discord

---

## Development

- GitHub
- GitLab
- Bitbucket
- Jira
- Linear

---

## Productivity

- Google Docs
- Google Drive
- Microsoft 365
- Notion
- Confluence

---

## CRM

- HubSpot
- Salesforce
- Zoho CRM

---

## Finance

- Stripe
- Paystack
- Flutterwave
- QuickBooks
- Xero

---

## Marketing

- Meta
- LinkedIn
- X
- Mailchimp
- Google Analytics

---

## Storage

- Google Drive
- Dropbox
- OneDrive

- Box

---

## Connector Architecture

Every connector contains:

- Identity
- Version
- Supported Events
- Supported Actions
- Required Permissions
- Authentication Method
- Configuration Schema
- Health Status

Connectors are plug-and-play.

---

## Connector Lifecycle

```
Install

↓

Configure

↓

Authenticate

↓

Validate

↓

Activate

↓

Monitor

↓
```

```
Update

↓

Disable

↓

Uninstall
```

Every connector follows the same lifecycle.

---

## Connector SDK

The HQ SDK allows developers to build connectors.

SDK modules include:

- Authentication
- Event Registration
- Action Registration
- Webhooks
- Logging
- Configuration
- Health Checks
- Error Handling

The SDK guarantees consistent integration quality.

---

## Integration Gateway

Responsibilities:

- Route requests
- Validate permissions
- Authenticate integrations
- Normalize responses
- Log activity
- Enforce rate limits

The gateway isolates HQ from external APIs.

---

## OAuth Manager

Supports:

- OAuth 2.0
- OpenID Connect
- API Keys
- Service Accounts
- Personal Access Tokens

Credentials are never exposed to executives.

---

## Secrets Vault

Stores:

- API Keys
- Refresh Tokens
- Client Secrets
- Certificates
- Webhook Secrets

Secrets are encrypted and rotated when possible.

---

## Event Engine

Supported events include:

External Event

↓

Event Validation

↓

Workflow Trigger

↓

Mission Creation

↓

Executive Assignment

↓

Mission Execution

↓

Notification

Examples:

- New GitHub issue
- New Stripe payment
- New Slack message
- New customer ticket

---

## Workflow Automation

Example:

New Customer Complaint

↓

Customer Success Director

↓

Operations Director

↓

CEO Review

↓

Response Generated

↓

Send Email

Automation combines integrations with executive intelligence.

---

## Bidirectional Synchronization

Supported capabilities:

- Import data
- Export data
- Real-time synchronization
- Conflict detection
- Version reconciliation

Synchronization is resilient to failures.

---

## Marketplace

The HQ Marketplace enables organizations to:

- Discover integrations
- Install connectors
- Update connectors
- Remove connectors
- View ratings
- Manage permissions

Marketplace packages are digitally signed.

---

# Connector Permissions

Each connector declares:

- Data access
- Required scopes
- Allowed actions
- Organization permissions

Users approve permissions before activation.

---

# Executive Integration

Executives can interact with external systems only through approved connectors.

Examples:

Technology Director

→ GitHub

Marketing Director

→ Meta Ads

Finance Director

→ Stripe

Sales Director

→ Salesforce

Customer Success Director

→ Zendesk

No executive communicates directly with third-party APIs.

---

## Integration Analytics

Track:

- Connector usage
- API latency
- Failure rate
- Active installations
- Event throughput
- Automation success rate

Analytics improve platform reliability.

---

## Security

The platform enforces:

- OAuth validation
- Encrypted credentials
- Rate limiting
- Audit logging
- Organization isolation
- Permission checks
- Webhook signature verification

Every integration is monitored continuously.

---

## Developer Platform

Developers can publish:

- Connectors
- Executive extensions
- Workflow templates
- Prompt packs
- Knowledge packs
- Analytics modules

Future certification ensures marketplace quality.

---

## Version Management

Every connector includes:

- Semantic version
- Changelog
- Compatibility information
- Upgrade path
- Deprecation policy

Organizations control update timing.

---

## Future Evolution

Supports:

- Enterprise private connectors
- Industry-specific marketplaces
- AI-powered connector generation
- Cross-organization collaboration
- Low-code integration builder
- Marketplace monetization
- Community-developed extensions

The platform is designed to evolve into a complete ecosystem.

---

## Success Criteria

The Integration Platform succeeds when:

- New connectors can be added without modifying the core platform.
- Executives interact with external systems securely.
- Organizations can automate workflows across applications.

- Developers can extend HQ through a stable SDK.
- The Marketplace becomes the central distribution channel for integrations and extensions.

---

## Conclusion

The HQ Integration Platform, SDK & Marketplace Architecture transforms HQ into an extensible AI Executive Operating System.

By combining a secure Integration Gateway, standardized Connector SDK, OAuth management, event-driven workflows, a developer ecosystem, and a Marketplace for extensions, HQ moves beyond a standalone product and becomes the intelligent coordination layer connecting organizations, AI executives, and the software they already rely on.